

This is the temporary name of the project we will develop over the semester.

I copy the notes created during our first session (02 Nov 2025) brainstorm.

This is a pre-draft version of the PRD for our project.

Vision

- Help Teachers and Students in a CS Dept at University better teach and better learn using AI.

Objectives

- The product is used by students of the CS Dept to learn the material of existing courses.
- Teachers in the CS Dept use the product to better teach existing courses.
- The product does not create new material, it does not replace existing AI tools (notebookLM, existing chatbots) - it enhances the learning experience around existing material.

Motivation

- Teachers worry students may use AI technology as a shortcut to avoid spending time on learning, and end-up not learning the material even though the students feel they can fulfill the course requirements easily.
- Students want to have enhanced access to course material to learn better, deeper, faster and in a more personalized manner.

Target Audience

- Teachers of Courses at the CIS Faculty @ BGU
- Students taking Courses in the Faculty

Features and Functionality

Overview of Students Features

- Chatbot on Course Material
- Access to Additional relevant resources (papers, videos, tutorials, exercises)
- Make using the course "platform" easier: install tools, debug issues, compile, submit assignments...
- Motivate: Why do I learn this material? Why is it important? What depends on it? Why is it beautiful?
- Generate personalized quizzes, exams, exercises to exercise the content and verify that the content is mastered. Check that the tasks are performed correctly.

Overview of Teacher Features

- Prepare content for the course material and make it accessible to students with AI
- Monitor students progress
- Prepare "learning trajectories" (sequence of topics, ordered by dependency, to be mastered by the students).
- Prepare assignments (create quizzes, assignments, exams, verify they are testable, correspond to the material taught, are "LLM resilient", are of the right length and complexity)
- Grade assignments
- Detect cheating

Student: Chatbot on Course Material

- Ensure the scope of the chat sessions is restricted to the course material
 - Filter out questions not related to the course
 - Adjust answers to the current level of the student as recorded in the system
- Provide a **socratic learning** experience
 - Do not answer questions directly - check misconceptions, analyze intentions, check perspectives behind the student questions before answering.
- Monitor Student Progress
 - Check what the student already knows and where the student is lacking knowledge (skills)
 - Keep track of what skills and content the student has acquired (progress on "learning trajectories").

Teacher: Prepare Content for the Course Chatbot

- Allow the teacher to upload relevant files about the course content (notes, documents, exams, slides, exercises, quizzes).
- Review the consistency of the course material.
- Index the course material to support a RAG approach.